



A Practical Journal of MODERN NETWORKING

A PRACTICAL JOURNAL SUBMIT TO THE

University of Mumbai



Submitted in partial fulfillment of the

Requirement for the award of the degree of

MASTER OF SCIENCE (INFORMATION TECHNOLOGY)

(SEMESTER 2) By

Full Name: NAZIFA SAJID HAJWANE

Roll No: 03

Under the Esteemed Guidance of

Mr. Raju Kukade

REENA MEHTA COLLEGE

(Affiliated to University of Mumbai)

THANE - 401101 - 2025 - 2026



**REENA MEHTA COLLEGE OF COMMERCE, ARTS, SCIENCE, AND MANAGEMENT
STUDIES**

ISO 9001-2015 CERTIFIED

**Reena Mehta College, Opp. Maxus Mall, Bhayandar (West).
Thane - 401101**

"MODERN NETWORKING"

CERTIFICATE

The Practical's Duly Signed In This Journal Represent the Bonafide Work Carried Out by
NAZIFA SAJID HAJWANE , ROLL NO: 03. A Student of Semester 2, FIRST YEAR MASTER
OF SCIENCE IN INFORMATION TECHNOLOGY (**M.SC. IT**), In the Computer Laboratory of
This College During the Academic Session **2025–2026.**

Lecturer In-Charge

Head Department of M.Sc. IT

External Examiner

Principal

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Practical No 1

Configure IP SLA Tracking and Path Control Topology

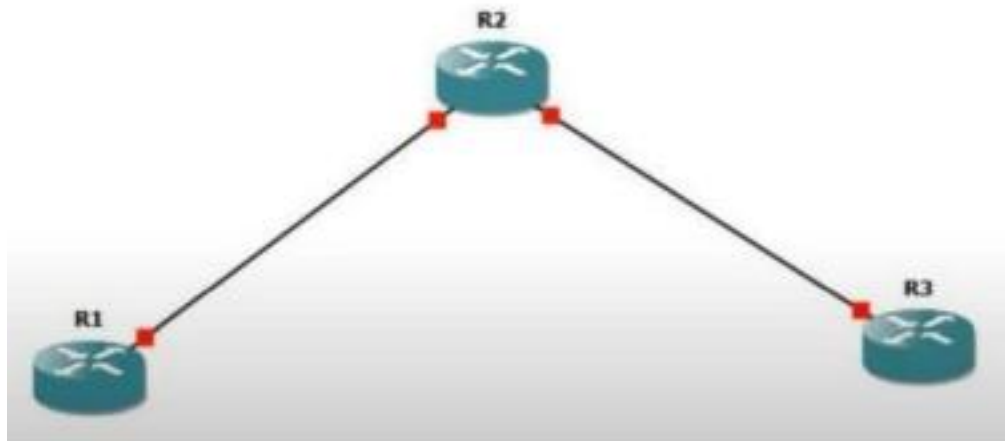
Aim: To configure and verify IP SLA (Service Level Agreement) tracking and path control on Cisco routers, enabling dynamic failover between multiple ISPs based on real-time network performance metrics.

Theory: IP SLA is a Cisco IOS feature that allows active monitoring of network performance by generating traffic and measuring parameters like delay, jitter, and packet loss. This proactive monitoring helps in assessing the quality of network paths.

Path Control involves directing traffic over specific network paths based on performance metrics. By integrating IP SLA with tracking objects, routers can make intelligent routing decisions, ensuring optimal path selection and network resilience.

Tracking Objects are used to monitor the status of IP SLA operations. If an IP SLA operation fails (e.g., due to high latency or packet loss), the tracking object reflects this state, allowing the router to adjust its routing decisions accordingly.

Topology:



Commands:

Configure IP SLA Operations

```
R1(config)# ip sla 1
```

```
R1(config-ip-sla)# icmp-echo 209.165.200.254 source-interface Serial0/0/0
```

```
R1(config-ip-sla-echo)# timeout 5000
```

```
R1(config-ip-sla-echo)# frequency 10
```

```
R1(config)# ip sla schedule 1 life forever start-time now
```

Explanation: This configuration sets up an ICMP echo operation (ping) to the IP

address 209.165.200.254, using Serial0/0/0 as the source interface. The operation has a timeout of 5000 ms and a frequency of 10 seconds.

Configure Tracking Objects

```
R1(config)# track 1 ip sla 1 reachability
```

```
R1(config-track)# delay down 10 up 1
```

```
R1(config-track)# exit
```

Explanation: This command exit creates a tracking object (ID 1) that monitors the reachability of IP SLA operation 1. The delay parameters ensure that transient failures don't cause unnecessary route changes.

Configure Static Routes with Tracking

```
R1(config)# ip route 0.0.0.0 0.0.0.0 209.165.201.1 5 track 1
```

```
R1(config)# ip route 0.0.0.0 0.0.0.0 209.165.202.129 10 track 2
```

Explanation: These commands set up two default routes with different administrative distances. The first route uses 209.165.201.1 with a distance of 5 and is tracked by object 1. The second route uses 209.165.202.129 with a distance of 10 and is tracked by object 2. Configure IP SLA Responder on Remote Router

```
R2(config)# ip sla responder
```

Explanation: This command enables the IP SLA responder on the remote router (R2), allowing it to respond to the ICMP echo requests sent by R1.

Verification:

To verify the IP SLA operation:

```
R1# show ip sla statistics
```

To check the status of tracking objects:

```
R1# show track
```

To view the routing table and confirm active routes:

```
R1# show ip route
```

Conclusion:

By configuring IP SLA tracking and path control, the network can dynamically adjust to changing conditions, ensuring optimal path selection and improved network resilience. This setup is particularly beneficial in scenarios where multiple ISPs are used, allowing for automatic failover in case of link degradation or failure.

Practical No. 2

Using the AS_PATH Attribute

Aim: To understand and demonstrate the use of the AS_PATH attribute in Border Gateway Protocol (BGP) for path selection and traffic engineering by simulating a multi-AS network topology using GNS3.

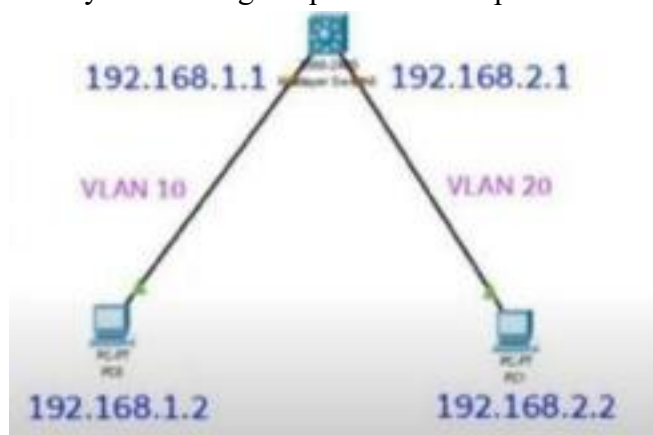
Theory: The Border Gateway Protocol (BGP) is a standardized exterior gateway protocol designed to exchange routing and reachability information among autonomous systems (ASes) on the Internet. BGP is classified as a path-vector routing protocol, and it makes routing decisions based on paths, network policies, or rule-sets configured by a network administrator.

One of the key attributes in BGP is the AS_PATH attribute. This attribute lists the sequence of ASes that routing information has traversed. It serves two primary purposes:

Loop Prevention: By examining the AS_PATH, a BGP router can detect routing loops. If a router sees its own AS number in the AS_PATH of a received route, it will reject that route to prevent a loop.

Path Selection: BGP prefers routes with shorter AS_PATHs, assuming all other attributes are equal. This preference helps in selecting the most efficient path to a destination.

Additionally, network administrators can manipulate the AS_PATH attribute using a technique called AS Path Prepending. This involves adding multiple instances of an AS number to the AS_PATH to make a route less attractive, thereby influencing the path selection process.



In the GNS3 simulation, we set up a network topology with three routers, each representing a different AS:

Router R1: AS 100 Router

R2: AS 200

Router R3: AS 300

The goal is to advertise a network from R1 and observe how AS_PATH influences the route selection on R3.

Step 1: Configure BGP on R1 router

```
bgp 100
```

```
network 192.168.1.0 mask 255.255.255.0 neighbour
```

```
10.0.12.2 remote-as 200
```

Step 2: Configure BGP on R2 router

```
bgp 200
```

```
neighbour 10.0.12.1 remote-as-100 neighbour
```

```
10.0.23.3 remote-as 300
```

Step 3: Configure BGP on R3

```
Router bgp 300
```

```
Neighbour 10.0.23.2 remote-as 200
```

Step 4: Verify AS_PATH on R3

On R3, use the following command to view the BGP

```
table: show ip bgp
```

We observe the route to 192.168.1.0/24 with an AS PATH of 200 100, indicating that the route has traversed AS 200 and AS 100.

Step 5: Apply AS Path Prepending on R2

To make the path through R2 less preferred, prepend AS 200 multiple times:

```
router bgp 200 neighbour 10.0.23.3 remote-as 300 neighbour 10.0.23.3 route-  
map PREPEND out route-map PREPEND permit 10 set
```

```
as-path prepend 200 200 200
```

After applying the route-map, R3 will see the AS_PATH as 200 200 200 100, making it longer and thus less preferred compared to other available paths.

Conclusion: This practical demonstrates the significance of the AS_PATH attribute in BGP for both loop prevention and path selection. By manipulating the AS_PATH using techniques like AS Path Prepending, network administrators can influence routing decisions to achieve desired traffic engineering outcomes. The GNS3 simulation effectively illustrates how BGP

routers use the AS_PATH attribute to determine the best path to a destination network.

Practical No. 3

Configuring IBGP and EBGP Sessions, Local Preference, and MED **Aim:** To configure IBGP, EBGP and set the local preference and MED

Theory: Border Gateway Protocol (BGP) Overview

BGP (Border Gateway Protocol) is a routing protocol used to exchange routing information between different networks on the internet. It is classified into two types:

External BGP (EBGP) - Used between different Autonomous Systems (AS).

Internal BGP (IBGP) - Used within the same Autonomous System.

External BGP (EBGP)

EBGP is used to exchange routes between different Autonomous Systems (ASes).

It is commonly used by Internet Service Providers (ISPs) and large enterprises to communicate with external networks.

The default Time-To-Live (TTL) value is 1, meaning that EBGP peers must be directly connected unless explicitly configured otherwise.

AS-path attribute is used in EBGP to prevent routing loops.

It prefers shorter AS paths when selecting the best route.

Example Scenario:

If AS100 wants to exchange routes with AS200, they establish an EBGP connection between their routers.

Internal BGP (IBGP)

IBGP is used for routing within the same Autonomous System.

It ensures that all routers in an AS have a consistent view of external routes learned via EBGP.

Unlike EBGP, IBGP does not modify the AS-path attribute.

IBGP requires a full mesh of connections (or Route Reflectors/Confederations to reduce overhead). Next-hop attribute must be reachable within the AS for proper routing.

Example Scenario:

If AS 100 has multiple routers, they must use IBGP to share routes learned from EBGP peers.

Key Differences Between EBGP and IBGP Feature EBGP IBGP

Used for Between different ASes Within the same AS AS-Path

Yes No

Modification

Next-Hop Yes No (next-hope must reachable) Change

Default TTL I 255

Full Mesh Required? No Yes (or use Route Reflectors)

MED and Local Preference in BGP

BGP (Border Gateway Protocol) uses several attributes to influence POP routing decisions. Two important attributes that help in path selection are MED (Multi-Exit Discriminator) and Local Preference.

Multi-Exit Discriminator (MED)

Purpose: MED is used to influence the incoming traffic from an external AS by suggesting the preferred entry point into an AS when multiple links exist.

Characteristics:

It is an optional, non-transitive attribute.

A lower MED value is preferred.

MED is shared only with directly connected external neighbors (not propagated beyond the next AS).

It is commonly used between ISPs or between enterprise networks and ISPs.

Example Scenario:

If AS100 has two links to AS200, it can set a lower MED on one link to tell AS200 to prefer that path for incoming traffic.

Local Preference

Purpose: Local Preference is used within an AS to influence the outgoing traffic by selecting the preferred exit point when multiple paths to the same destination exist.

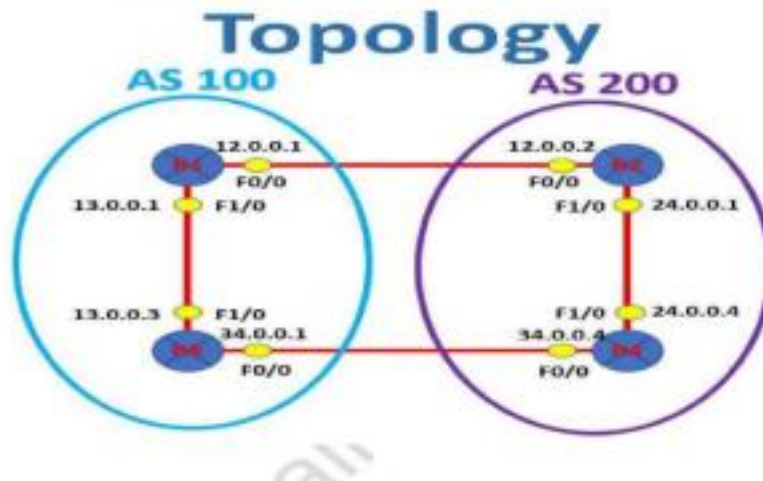
Characteristics:

- It is a well-known, discretionary attribute.
- A higher Local Preference value is preferred.

- It is propagated within the AS to all IBGP peers.
 - Used mainly by ISPs and large networks to control outbound traffic flow.
- Example Scenario:

If AS100 has two exit points (R1 and R2) to AS200, setting a higher Local Preference on R1 will make all routers in AS100 prefer R1 for outgoing traffic.

We use the following topology



We do the configuration using the following steps

Step 1: Configure the IP addresses on all the Routers

Router 1

R1#

R1#configure terminal

R1(config)#

R1(config)#interface fastEthernet 0/0

R1(config-if)#

R1(config-if)#ip address 12.0.0.1 255.255.255.0

R1(config-if)#

R1(config-if)#no shutdown

R1(config-if)#

R1(config-if)#exit

R1(config)#

R1(config)#interface fastEthernet 1/0

R1(config-if)#ip address 13.0.0.1 255.255.255.0

R1(config-if)#no shutdown

R1(config-if)#

R1(config-if)#exit

R1(config)

Router 2

R2#

R2#configure terminal

Enter configuration commands, one per line. End with CNTL/Z.

R2(config)#

R2(config)#interface fastEthernet 0/0

R2(config-if)#

R2(config-if)#ip address 12.0.0.2 255.255.255.0

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#

R2(config)#interface fastEthernet 1/0

R2(config-if)#

R2(config-if)#ip address 24.0.0.1 255.255.255.0

R2(config-if)#no shutdown

R2(config-if)#exit

R2(config)#

Router 3

R3#

R3#configure terminal

R3(config)#

R3(config)#interface fastEthernet 0/0

R3(config-if)#

R3(config-if)#ip address 34.0.0.1 255.255.255.0

R3(config-if)#no shutdown

R3(config-if)#exit

R3(config)#

R3(config)#interface fastEthernet 1/0

R3(config-if)#

R3(config-if)#ip address 13.0.0.3 255.255.255.0

R3(config-if)#no shutdown

R3(config-if)#exit

R3(config)#

Router4

R4#

R4#configure terminal

```
R4(config)#
R4(config)#interface fastEthernet 0/0
R4(config-if)#
R4(config-if)#ip address 34.0.0.4 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
R4(config)#interface fastEthernet 1/0
R4(config-if)#
R4(config-if)#ip address 24.0.0.4 255.255.255.0
R4(config-if)#no shutdown
R4(config-if)#exit
R4(config)#
```

Step 2: Set IBGP and EBGp on each router

Router 1

```
R1(config)#router bgp 100
R1(config-router)#
R1(config-router)#neighbor 13.0.0.3 remote-as 100
R1(config-router)#neighbor 13.0.0.3 update-source fastEthernet 1/0
R1(config-router)# network 12.0.0.0 mask 255.255.255.0 R1(config-
router)#neighbor 12.0.0.2 remote-as 200 R1(config-router)#network 13.0.0.0
mask 255.255.255.0 R1(config-router)#
R1(config-router)#exit
R1(config)#exit
R1#
```

Router 2

```
R2(config)#router bgp 200
R2(config-router)#
R2(config-router)#neighbor 24.0.0.4 remote-as 200 R2(config-
router)#neighbor 24.0.0.4 update-source fastEthernet 1/0 R2(config-
router)#network 12.0.0.0 mask 255.255.255.0 R2(config-router)#neighbor
12.0.0.1 remote-as 100 R2(config-router)#network 24.0.0.0 mask
255.255.255.0 R2(config-router)#exit
R2(config)#
```

Router 3

```
R3(config)#router bgp 100
R3(config-router)#
R3(config-router)#neighbour 13.0.0.1 remote-as 100 R3(config-
router)#neighbour 13.0.0.1 update-source fastEthernet 1/0 R3(config-
router)#network 34.0.0.0 mask 255.255.255.0 R3(config-router)#neighbor
```

```
34.0.0.4 remote-as 200 R3(config-router)#network 13.0.0.0 mask
255.255.255.0 R3(config-router)#
```

Router 4

```
R4(config)#
R4(config)#router bgp 200
R4(config-router)#
R4(config-router)#neighbor 24.0.0.1 remote-as 200 R4(config-
router)#neighbor 24.0.0.1 update-source fastEthernet 1/0 R4(config-
router)#network 34.0.0.0 mask 255.255.255.0
R4(config-router)#neighbor 34.0.0.3 remote-as 100
R4(config-router)#network 24.0.0.0 mask 255.255.255.0
R4(config-router)#exit
R4(config)#
```

Step 4: Verify the BGP protocol by pinging from Router1 to all interfaces

```
R1#ping 12.0.0.2
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 12.0.0.2, timeout is 2 seconds: !!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 20/29/36 ms

```
R1#ping 13.0.0.3
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 13.0.0.3, timeout is 2 seconds: !!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 16/26/36 ms

```
R1#ping 24.0.0.4
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 24.0.0.4, timeout is 2 seconds: !!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 56/60/64 ms

```
R1#ping 34.0.0.4
```

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 34.0.0.4, timeout is 2 seconds: !!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 52/56/60 ms

Step 5: Configure Local Preference (influencing Outbound Traffic from AS 100)

```
R1#configure terminal
```

```
R1(config)#
```

```
R1(config)#route-map smile_ip permit 10
```

```
R1(config-route-map)#
```

```
R1(config-route-map)#set local-preference 200
```

```
R1(config-route-map)#
```

```

R1(config-route-map)#exit
R1(config)#router bgp 100
R1(config-router)#
R1(config-router)#neighbor 13.0.0.3 route-map smile_ip in
R1(config-router)#exit
R1(config)#exit

```

Step 6: Configure MED (Influencing Inbound Traffic to AS 100)

```

R2(config)#
R2(config)#route-map set_med permit 10
R2(config-route-map)#exit
R2(config)#
R2(config)#router bgp 200
R2 (config-router)#
R2(config-router)#neighbor 12.0.0.1 route-map set_med out
R2(config-router)#exit
R2(config)#

```

Step 7: Verification

```

R1#
R1#show ip bgp summary
BGP router identifier 13.0.0.1, local AS number 100  BGP
table version is 7, main routing table version 7  4 network
entries using 576 bytes of memory
6 path entries using 480 bytes of memory
3/3 BGP path/bestpath attribute entries using 408 bytes of memory  1 BGP
AS-PATH entries using 24 bytes of memory  0 BGP route-map cache entries
using 0 bytes of memory  0 BGP filter-list cache entries using 0 bytes of
memory  BGP using 1488 total bytes of memory
BGP activity 4/0 prefixes, 6/0 paths, scan interval 60 secs

```

```

R1#show ip route
Codes: L - local, C - connected, S - static, R- RIP, M- mobile, B - BGP  D -
EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  N1- OSPF NSSA
external type 1, N2 - OSPF NSSA external type 2  E1 - OSPF external type 1, E2
- OSPF external type 2  i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-
IS level-2  ia - IS-IS inter area, * - candidate default, U - per-user static route  o-
ODR, P - periodic downloaded static route, H - NHRP, I - LISP  + - replicated
route, % - next hop override

```

Gateway of last resort is not set

12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks C 12.0.0.0/24 is directly connected, FastEthernet0/0 L 12.0.0.1/32 is directly connected, FastEthernet0/0 13.0.0.0/8 is variably subnetted, 2 subnets, 2 masks C 13.0.0.1/32 13.0.0.0/24 is directly connected, FastEthernet1/0 L 13.0.0.1/32 is directly connected, FastEthernet1/0 24.0.0.0/24 is subnetted, 1 subnets B 24.0.0.0 [20/50] via 12.0.0.2, 00:00:24 34.0.0.0/24 is subnetted, 1 subnets B 34.0.0.0 [200/0] via 13.0.0.3, 00:07:31 R1#show ip bgp BGP table version is 7, local router ID is 13.0.0.1 Status codes: s suppressed, d damped, h history, * valid, > best, i - internal, r RIB-failure, S Stale, m multipath, b backup-path, f RT-Filter, x best-external, a additional-path, c RIB-compressed, Origin codes: i- IGP, e - EGP, ? - incomplete RPKI validation codes: V valid, I invalid, N Not found

```

Network Next Hop Metric LocPrf Weight Path
r> 12.0.0.0/24 12.0.0.2 50 0 200 i * i 13.0.0.0/24 13.0.0.3 0 100
0i *> 0.0.0.0 0 32768 i *> 24.0.0.0/24 12.0.0.2 50 0 200 i *
34.0.0.0/24 12.0.0.2 50 0 200 i *>i 13.0.0.3 0 100 0i
R1#show ip bgp neighbors
BGP neighbor is 12.0.0.2, remote AS 200, external link
BGP version 4, remote router ID 24.0.0.1
BGP state = Established, up for 00:09:48
Last read 00:00:16, last write 00:00:23, hold time is 180, keepalive
interval is 60 seconds
Neighbor sessions:
1 active, is not multisession capable (disabled)
Neighbor capabilities:
Route refresh: advertised and received(new)
Four-octets ASN Capability: advertised and received
Address family IPv4 Unicast: advertised and received
Enhanced Refresh Capability: advertised and received
Multisession Capability:
Stateful switchover support enabled: NO for session 1
Message statistics:
InQ depth is 0
OutQ depth is 0
Sent Rcvd
Opens: 1 1
Notifications: 0 0 Updates: 3 7
Keepalives: 12 12

```


Practical No 4

Secure the Management Plane

Aim: To Secure the management plane using various strategies

Theory: The management plane is responsible for managing a network device - such as configuring settings, monitoring status, and maintaining security. It allows administrators to communicate with the router or switch through command-line interfaces (CLI) or graphical interfaces.

Examples of management plane protocols:

- Telnet (insecure, plaintext)
- SSH (secure)
- SNMP
- HTTP/HTTPS
- Console/VTY access

The management plane is a critical attack surface. If compromised, attackers can:

- View or change device configurations
- Shut down interfaces
- Redirect traffic
- Launch attacks on other devices

Hence, securing it is essential for protecting the entire network infrastructure.

Methods to Secure the Management Plane

1. Use Secure Access Protocols
Replace insecure protocols like Telnet with SSH, which provides encrypted access to the CLI.
2. Implement User Authentication
Configure local usernames and passwords, and set privilege levels to limit what users can do.
3. Use Access Control Lists (ACLs)
Limit access to management interfaces by allowing only authorized IP addresses to connect.
4. Configure Strong Passwords & Encryption
Use 'enable secret, service password-encryption', and avoid weak or default credentials.
5. Disable Unused Services
Turn off services like CDP, HTTP, FTP, and bootp that are not in use to

reduce the attack surface.

6. Display Banner Warnings

Configure 'banner motd' and 'banner login' to display legal notices or warnings to deter unauthorized users.

7. Verify and Test the Setup

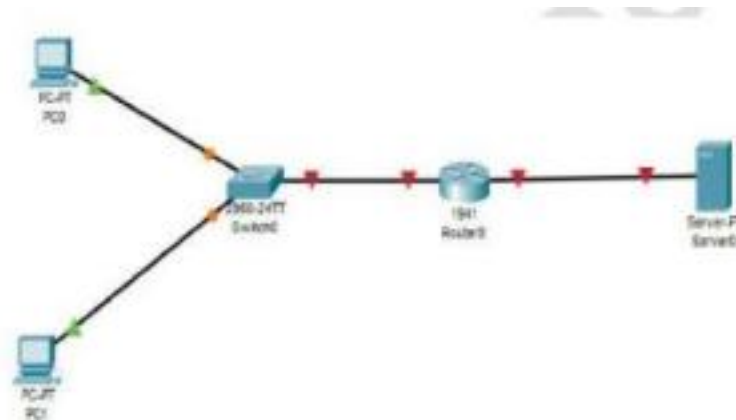
Always test your configuration using commands like 'show running config', 'show ip ssh', 'show access-lists', and attempt authorized and unauthorized access.

Real-World Importance

In enterprise networks, securing the management plane ensures: - Only authorized personnel can make changes

- Devices are not compromised remotely
- Regulatory compliance with security standards (like ISO, NIST, etc.) - Reduced risk of configuration tampering, insider threats, and cyber attacks.

Topology:



We use the following topology

IP Addressing Table:

Device	Interface	IP Address	Subnet Mask	Gateway
PC0	FastEthernet0	192.168.1.2	255.255.255.0	192.168.1.1
PC1	FastEthernet0	192.168.1.3	255.255.255.0	192.168.1.1
Router	GEthernet0/0	192.168.1.1	255.255.255.0	
Router	GEthernet0/1	192.168.2.1	255.255.255.0	
Server	FastEthernet0	192.168.2.2	255.255.255.0	192.168.2.1

We Configure the IP addresses as follows

PC0:

The screenshot shows the configuration window for PC0. The 'Config' tab is active, and the 'IP Configuration' section is expanded. The 'Interface' is set to 'FastEthernet0/0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 settings are: IP Address: 192.168.1.2, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1, and DNS Server: 8.8.8.8. The IPv6 section shows 'Static' selected with an empty IP Address field, a Link Local Address of FE80::280:2FFF:FEA3::C8EA, and empty fields for Default Gateway and DNS Server. The '802.1X' section has 'Use 802.1X Security' unchecked, and the 'Authentication' dropdown is set to 'RADIUS' with empty fields for Username and Password. A 'Top' button is at the bottom left.

PC1

The screenshot shows the configuration window for PC1. The 'Config' tab is active, and the 'IP Configuration' section is expanded. The 'Interface' is set to 'FastEthernet0/0'. Under 'IP Configuration', the 'Static' radio button is selected. The IPv4 settings are: IP Address: 192.168.1.3, Subnet Mask: 255.255.255.0, Default Gateway: 192.168.1.1, and DNS Server: 8.8.8.8. The IPv6 section shows 'Static' selected with an empty IP Address field, a Link Local Address of FE80::280:2FFF:FEA3::C8EA, and empty fields for Default Gateway and DNS Server. The '802.1X' section has 'Use 802.1X Security' unchecked, and the 'Authentication' dropdown is set to 'RADIUS' with empty fields for Username and Password. A 'Top' button is at the bottom left.

Server:

Server0

Physical Config Services **Desktop** Programming Attributes

Configuration

IP Configuration

☐ DHCP ☒ Static

IPv4 Address: 192.168.2.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.2.1

DNS Server: 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address: /

Link Local Address: FE80:201:C7FF:FE48:800B

Default Gateway:

DNS Server:

802.1X

☐ Use 802.1X Security

Authentication: RADIUS

Username:

Password:

☐ Tip

Router0: interface GigabitEthernet0/0

Router0

Physical **Config** Attributes

GLOBAL

Settings

Algorithm Settings

ROUTING

Static

RIP

SWITCHING

VLAN Database

INTERFACE

GigabitEthernet0/0

GigabitEthernet0/1

GigabitEthernet0/0

Port Status: ☒ On

Bandwidth: 1000 Mbps 100 Mbps 10 Mbps ☒ Auto

Duplex: Half Duplex Full Duplex ☒ Auto

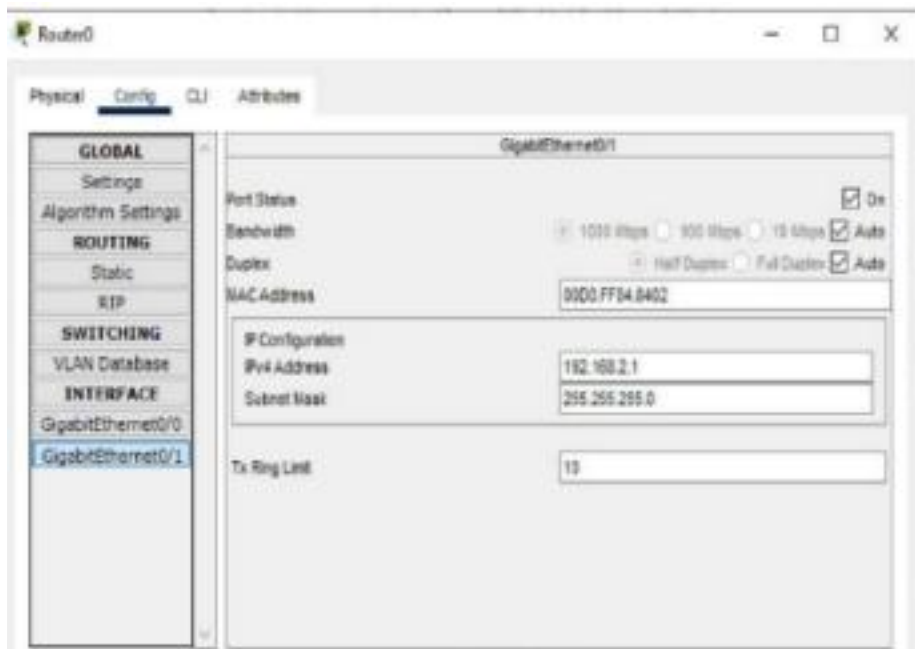
MAC Address: 0800.FF94.8401

IP Configuration

IPv4 Address: 192.168.1.1

Subnet Mask: 255.255.255.0

Tx Req Limit: 10



Now we secure the management plane using the following steps

Step 1: Secure access to Router0 via SSH

```
Router>enable
```

```
Router#
```

```
Router#configure terminal
```

```
Router(config)#
```

```
Router(config)#hostname R1
```

```
R1(config)#
```

```
R1(config)#ip domain-name
```

```
ismileacademy.com R1(config)#
```

```
R1(config)#crypto key generate rsa
```

The name for the keys will be: R1.ismileacademy.com

Choose the size of the key modulus in the range of 360 to 2048 for your General Purpose Keys.

Choosing a key modulus greater than 512 may take a few minutes.

How many bits in the modulus [512]: 2048 %

Generating 2048 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#ip ssh version 2

*Mar 1 0:25:48.611: %SSH-5-ENABLED: SSH 1.99 has been enabled

R1(config)#

R1(config)#ip ssh time-out 60

R1(config)#ip ssh authentication-retries 2

R1(config)#username admin privilege 15 secret smilehp

R1(config)#line vty 0 4

R1(config-line)#

R1(config-line)#transport input ssh

R1(config-line)#

R1(config-line)#login local

R1(config-line)#exit

R1(config)#

Step 2: Restrict Access to Router using access control list

R1(config)#access-list 10 permit 192.168.1.2

R1(config)#access-list 10 deny any

R1(config)#line vty 0 4

R1(config-line)#

R1(config-line)#access-class 10 in

R1(config-line)#exit

R1(config)#

R1(config)#interface gigabitEthernet 0/0

```
R1(config-if)#
```

```
R1(config-if)#ip access-group 10 in
```

```
R1(config-if)#exit
```

```
R1(config)#
```

Step 3: Disable unused services in the Server

```
R1(config)#
```

```
R1(config)#ip access-list extended smile
```

```
R1(config-ext-nacl)#
```

```
  R1(config-ext-nacl)#permit icmp host 192.168.1.2 host 192.168.2.2
```

```
  R1(config-ext-nacl)#permit tcp host 192.168.1.2 host 192.168.2.2 eq 21
```

```
  R1(config-ext-nacl)#permit tcp host 192.168.1.2 host 192.168.2.2 eq 22
```

```
  R1(config-ext-nacl)#deny ip any host 192.168.2.2
```

```
  R1(config-ext-nacl)#exit
```

```
  R1(config)#
```

```
  R1(config)#interface gigabitEthernet 0/1
```

```
    R1(config-if)#
```

```
    R1(config-if)#ip access-group smile out
```

```
    R1(config-if)#exit
```

```
    R1(config)#
```

Step 4: Configure Banner messages

```
R1(config)#banner motd b
```

```
Enter TEXT message. End with the character 'b'.
```

```
*****
```

```
* Unauthorized access is prohibited. Disconnect now!*
```

```
R1(config)#
```

```
R1(config)#banner login b
```

Enter TEXT message. End with the character 'b'.

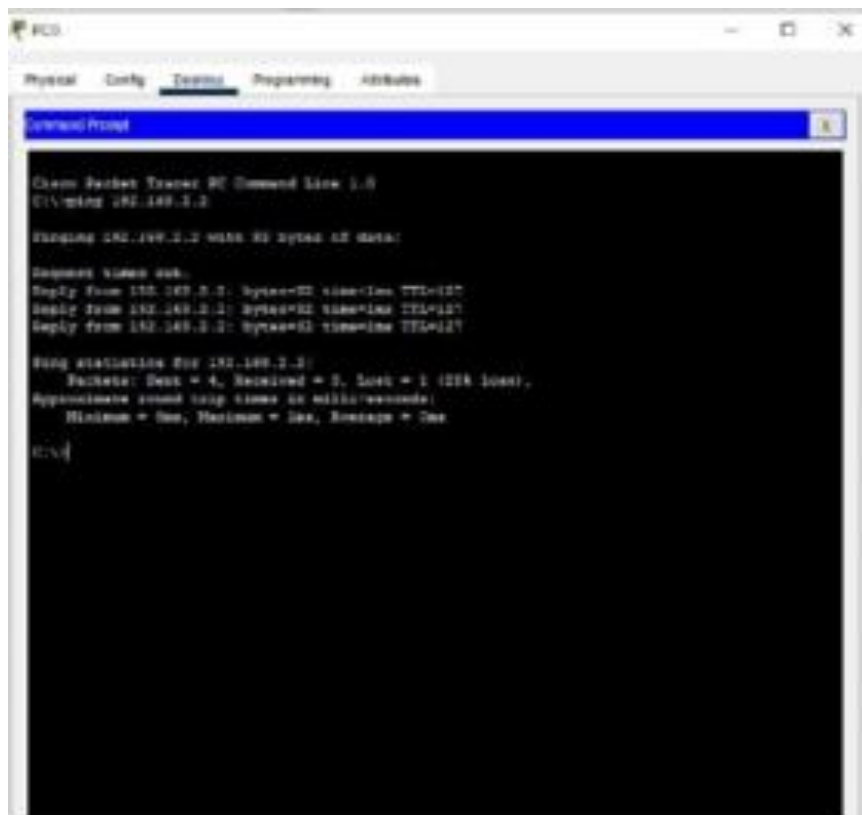
* Authorized personnel only. Unauthorized users beware!*

The above steps complete the configuration part of Secure management plane, now we verify it as follows

Output:

1) Checking for PING message

PC0: Checking for ping message from PC0 to Server



```
PC0
Physical Config Terminal Programming Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data:

Request timeout.
Reply from 192.168.1.2: bytes=32 time=1ms TTL=127
Reply from 192.168.1.2: bytes=32 time=1ms TTL=127
Reply from 192.168.1.2: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.1.2:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

It is successful

PC1: Checking for ping message from PC1 to Server


```
Physical  Config  Command  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: Destination host unreachable.
Reply from 192.168.2.2: Destination host unreachable.
Reply from 192.168.2.2: Destination host unreachable.
Reply from 192.168.2.2: Destination host unreachable.

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
C:\>
```

It is failure

2) Checking for remote login using ssh

PC0:

```
Physical  Config  Command  Programming  Attributes

Command Prompt

Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.2.2

Pinging 192.168.2.2 with 32 bytes of data:

Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127
Reply from 192.168.2.2: bytes=32 time=1ms TTL=127

Ping statistics for 192.168.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milliseconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\>ssh -S admin 192.168.2.1

Password:
.....
* Unauthorized access is prohibited
exit
```

It is success

PC1:

3) Checking for ftp

PC0:

It is success

PC1:

It is failure

4) Checking for http:
PC0:

It does not connect and times out

PC1:

It does not connect and times out

The following things are configured while performing the Secure management plane also verified

Test	PCI (192.168.1.2)	PC2(192.168.1.3)
Ping server	Allowed	Blocked
FTP to server	Allowed	Blocked
SSH to server	Allowed	Blocked
HTTP/HTTPS/DNS etc.	Blocked	Blocked

Practical No 5

Configure and Verify Path Control Using PBR

Aim: To configure and verify path control using Policy Based Routing(PBR)

Theory:

In traditional IP routing, routers forward packets based only on the destination IP address, using the longest prefix match from their routing table.

Policy-Based Routing (PBR) allows you to override this behavior by defining custom routing policies based on other criteria like:

- 1) Source IP address
- 2) Protocol
- 3) Packet size
- 4) Incoming interface

This gives network administrators greater flexibility in managing traffic flow based on business or technical policies.

PBR is commonly used when you need to:

- 1) Route specific users or devices through different ISPs or firewalls
- 2) Apply different Quality of Service (QoS) or bandwidth policies to selected traffic
- 3) Bypass certain links for sensitive data
- 4) Perform load balancing or failover routing between multiple paths

PBR uses three main configuration components on a router:

- 1) Access Control List (ACL): Used to match the traffic that should be treated differently (e.g., match source IPs).
- 2) Route Map: Defines the policy for matched traffic, such as setting a specific next-hop IP.
- 3) IP Policy Statement: Applied to an interface using the ip policy route-map command, telling the router to evaluate and apply the route map to incoming traffic on that interface.

Example Use Case

You have two exit routers (R2 and R3).

You want a specific PC (e.g., 192.168.1.2) to reach the internet only through R2, even though R3 is the default route.

PBR allows you to match traffic from that PC and force it through R2, regardless of the routing table.

Benefits of PBR

- 1) Granular control over traffic flows
- 2) Can enforce security, compliance, and routing policies
- 3) Useful in multi-homed networks (with multiple ISPs)

Limitations of PBR

- 1) It only applies to incoming packets on the interface where the policy is configured.
- 2) Requires careful design to avoid routing loops or black holes
- 3) Adds some processing overhead on routers (not recommended on low-end devices for high-volume traffic)

We use the following topology:

We do the configuration using the following steps **Step 1:**

Configure the IP addresses on all the Routers and PCs

Router 1

```
R1#
```

```
R1#configure terminal
```

```
R1(config)#
```

```
R1(config)#interface fastEthernet 0/0
```

```
R1(config-if)#
```

```
R1(config-if)#ip address 192.168.1.1
```

```
255.255.255.0 R1(config-if)#no shutdown
```

```
R1(config-if)#exit
```

```
R1(config)#  
R1(config)#interface fastEthernet 1/0  
R1(config-if)#  
R1(config-if)#ip address 10.0.0.1 255.0.0.0  
R1(config-if)#no shutdown  
R1(config-if)#exit  
R1(config)#  
R1(config)#interface fastEthernet 2/0  
R1(config-if)#  
R1(config-if)#ip address 20.0.0.1 255.0.0.0  
R1(config-if)#no shutdown  
R1(config-if)#exit  
R1(config)#
```

Router 2

```
R2#  
R2#configure terminal  
R2(config)#  
R2(config)#interface fastEthernet 0/0  
R2(config-if)#  
R2(config-if)#ip address 192.168.2.1  
255.255.255.0 R2(config-if)#no shutdown  
R2(config-if)#exit  
R2(config)#  
R2(config)#interface fastEthernet 1/0
```

```
R2(config-if)#  
  
R2(config-if)#ip address 10.0.0.2  
255.0.0.0 R2(config-if)#no shutdown  
  
R2(config-if)#exit  
  
R2(config)#  
  
R2(config)#interface fastEthernet 2/0  
  
R2(config-if)#  
  
R2(config-if)#ip address 30.0.0.1 255.0.0.0  
R2(config-if)#no shutdown  
  
R2(config-if)#exit  
  
R2(config)#
```

Router 3

```
R3#  
  
R3#configure terminal  
  
R3(config)#  
  
R3(config)#interface fastEthernet 0/0  
  
R3(config-if)#  
  
R3(config-if)#ip address 192.168.3.1  
255.255.255.0 R3(config-if)#no shutdown  
  
R3(config-if)#exit  
  
R3(config)#  
  
R3(config)#interface fastEthernet 2/0  
  
R3(config-if)#  
  
R3(config-if)#ip address 30.0.0.2
```

255.0.0.0 R3(config-if)#no shutdown

R3(config-if)#exit

R3(config)#

R3(config)#interface fastEthernet 1/0

R3(config-if)#

R3(config-if)#ip address 20.0.0.2 255.0.0.0

R3(config-if)#no shutdown

R3(config-if)#exit

R3(config)#

Configuring PC1

PC1>ip 192.168.1.2 255.255.255.0 192.168.1.1

Configuring PC2

PC2>ip 192.168.2.2 255.255.255.0 192.168.2.1

Configuring PC3

PC3>ip 192.168.3.2 255.255.255.0 192.168.3.1

Step 2: Checking the connectivity between the networks using ping

R1#ping 192.168.2.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.2.2, timeout is 2 seconds:

Success rate is 0 percent (0/5)

R1#ping 192.168.3.2

Type escape sequence to abort.

Sending 5, 100-byte ICMP Echos to 192.168.3.2, timeout is 2 seconds:

Success rate is 0 percent (0/5)

R1#ping 192.168.1.2

Type Sending escape 5, 100-byte sequence ICMP to abort.

Sending 5, 100-byte ICMP Echos to 192.168.1.2, timeout is 2 seconds:

!!!!

Success rate is 100 percent (5/5), round-trip min/avg/max = 48/63/88 ms

Step 5: Checking the path before setting Policy Based Routing (PBR) Now we trace the route the packets traverse when passing from PC1 to PC2 and PC1 to PC3

From PC1 to PC2

PC1> trace 192.168.2.2

trace to 192.168.2.2, 8 hops max, press Ctrl+C to stop

1 192.168.1.1 35.841 ms 4.254 ms 23.860 ms

2 10.0.0.2 130.033 ms 66.677 ms 72.610 ms

3 * **

4 *192.168.2.2 75.028 ms (ICMP type:3, code:3, Destination port unreachable)

The above route can be visualized as (shown by red lines)

The given outputs are obvious as the best possible routes from PC1 To PC2 and also for PC1 to PC3

Step 6: Setting Policy Based Routing (PBR) in

Router1

R1#

R1#configure terminal

Enter configuration commands, one per line. End with

CNTL/Z. R1(config)#


```
R1(config)#access-list 10 permit 192.168.1.2
```

```
R1(config)#route-map SMILE permit 10
```

```
R1(config-route-map)#
```

```
R1(config-route-map)#match ip address 10
```

```
R1(config-route-map)#set ip next-hop 10.0.0.2
```

```
R1(config-route-map)#exit
```

```
R1(config)#
```

```
R1(config)#interface fastEthernet 0/0
```

```
R1(config-if)#
```

```
R1(config-if)#ip policy route-map SMILE
```

```
R1(config-if)#exit
```

```
R1(config)#
```

Step 7: Verifying the route from PC1 to PC3

```
PC1> trace 192.168.3.2
```

```
trace to 192.168.3.2, 8 hops max, press Ctrl+C to stop
```

```
1 192.168.1.1 9.682 ms 30.054 ms 20.347 ms
```

```
2 10.0.0.2 79.511 ms 70.173 ms 89.750 ms
```

```
3 30.0.0.2 100.528 ms 80.136 ms 109.753 ms
```

```
4 * **
```

```
5 *192.168.3.2 160.514 ms (ICMP type:3
```

The above route can be visualized as (shown by red lines)

The packets from PC1 to PC3 are not forwarded directly from R1 to R3 but they take a long route due to the Policy Based Routing, they are forwarded from PC1 to R1 and from R1 to R2 and then from R2 to R3 and finally to PC3

Practical No 6

IP SLA and Remote SPAN in a Campus Environment

Aim: To demonstrate the configuration and use of IP SLA and Remote SPAN (RSPAN) in a campus environment for performance monitoring and traffic analysis.

Theory: 1. IP SLA (IP Service Level Agreement) IP SLA is a Cisco feature used to measure network performance and monitor service levels by generating synthetic traffic between network devices. It helps in:

Measuring latency, jitter, packet loss, and availability

Verifying QoS (Quality of Service)

Troubleshooting network paths

Tracking reachability to trigger dynamic routing changes

How it works: IP SLA sends test traffic (like ICMP, UDP, or HTTP) to a target IP address.

It records statistics like round-trip time and availability.

You can schedule these tests and track trends over time.

Remote SPAN (RSPAN)

Remote SPAN extends the Switched Port Analyzer (SPAN) feature to span traffic across

Multiple switches in a campus environment. It is used to:

Capture packets from a source port/VLAN to a central analyzer

Troubleshoot performance or security issues centrally

How it works:

You configure a special RSPAN VLAN to carry mirrored traffic. Source switches send mirrored traffic to this VLAN.

The destination switch receives mirrored traffic on an RSPAN port where an analyzer or packet capture tool is connected.

Code:

1. IP SLA Configuration Example (Cisco CLI)

```
conf t
ip sla 1
icmp-echo 192.168.1.1
frequency 10
exit
ip sla schedule 1 life forever start-time now
track 1 ip sla 1 reachability
```

This config pings 192.168.1.1 every 10 seconds and tracks its availability.

2. Remote SPAN Configuration Example

On the Source Switch:

```
conf t
vlan 999
remote-span
exit
monitor session 1 destination source interface FastEthernet0/1
monitor session 1 destination remote vlan 999
```

On the Destination Switch:

```
conf t
vlan 999
remote-span
exit
monitor session 1 source remote vlan 999
monitor session 1 destination interface FastEthernet0/24
```

Connect a Wireshark PC or analyzer tool to Fa0/24 on the destination switch to capture traffic

Practical No 7

Inter-VLAN Routing

Aim: To Study the Inter-VLAN routing on a layer-3 switch

Theory: VLANs are used to segment switched Layer 2 networks for a variety of reasons. Regardless of the reason, hosts in one VLAN cannot communicate with hosts in another VLAN unless there is a router or a Layer 3 switch to provide routing services.

Inter-VLAN routing is the process of forwarding network traffic from one VLAN to another VLAN.

There are three inter-VLAN routing options:

- 1) Legacy Inter-VLAN routing: This is a legacy solution. It does not scale well.
- 2) Router-on-a-Stick: This is an acceptable solution for a small- to medium-sized network.
- 3) Layer 3 switch using switched virtual interfaces (SVIs): This is the most scalable solution for medium to large organizations.

The modern method of performing inter-VLAN routing is to use Layer 3 switches and switched virtual interfaces (SVI). An SVI is a virtual interface that is configured on a Layer 3 switch

Inter-VLAN SVIs are created the same way that the management VLAN interface is configured. The SVI is created for a VLAN that exists on the switch. Although virtual, the SVI performs the same functions for the VLAN as a router interface would. Specifically, it provides Layer 3 processing for packets that are sent to or from all switch ports associated with that VLAN.

The following are advantages of using Layer 3 switches for inter-VLAN routing: 1) They are much faster than router-on-a-stick because everything is hardware switched and routed.

2) There is no need for external links from the switch to the router for routing. 3) They are not limited to one link because Layer 2 Ether Channels can be used as trunk links between the switches to increase bandwidth.

4) Latency is much lower because data does not need to leave the switch to be routed to a different network.

5) They are more commonly deployed in a campus LAN than routers. The only disadvantage is that Layer 3 switches are more expensive than Layer 2

switches, but they can be less expensive than a separate Layer 2 switch and router.

We use the following topology to study Inter-VLAN routing

We configure the IP addresses on the PC

PC0:

Now we configure the Multilayer switch using the following command in the CLI mode

```
Switch>enable
```

```
Switch#configure terminal
```

Enter configuration commands, one per line. End with

```
CNTL/Z. Switch(config)#vlan 10
```

```
Switch(config-vlan)#name smile
```

```
Switch(config-vlan)#exit
```

```
Switch(config)#
```

```
Switch(config)#vlan 20
```

```
Switch(config-vlan)#name cisco
```

```
Switch(config-vlan)#exit
```

```
Switch(config)#
```

```
Switch(config)#interface vlan 10
```

```
Switch(config-if)#
```

```
Switch(config-if)#ip address 192.168.1.1
```

```
255.255.255.0 Switch(config-if)#no shutdown
```

```
Switch(config-if)#exit
```

```
Switch(config)#
```

```
Switch(config)#interface vlan 20
```

```
Switch(config-if)#
```

```
Switch(config-if)#ip address 192.168.2.1
255.255.255.0 Switch(config-if)#no shutdown
Switch(config-if)#exit
Switch(config)#exit
Switch#
Switch#show ip interface brief
Switch(config)#ip routing
```

Output:

Now we ping PC1 from PC0 to check the connectivity

Practical No 8

Simulating OpenFlow Using Mininet

Aim: To simulate a Software Defined Networking (SDN) environment using Mininet and observe the working of the Open Flow protocol between a switch and a controller.

Theory: Software Defined Networking (SDN) is a modern network architecture that separates the control plane (decision-making) from the data plane (packet forwarding). This is done using a central controller that communicates with network switches via a protocol like OpenFlow.

Mininet is a lightweight network emulator that creates a virtual network with hosts, switches, and controllers. It uses Open vSwitch (OVS) to simulate software-defined switches.

In an OpenFlow network:

Switches send unknown requests with packet flow events rules to the controller.

Rules are installed into the switch's flow table.

Future packets matching those rules are forwarded directly by the switch without involving the controller again.

This practical demonstrates the basic operation of OpenFlow using a simple topology in Mininet.

Understanding Software Defined Networking (SDN) Traditional networks are often rigid and hardware-driven, where each switch or router has its own control logic. In contrast, Software Defined Networking (SDN) is a modern approach where the control plane (decision-making) is separated from the data plane (packet forwarding). This architecture gives network administrators centralized control and greater flexibility.

In SDN, the controller is the brain of the network. It decides how traffic should flow and instructs the switches accordingly using a standardized protocol - typically OpenFlow.

What is Open Flow?

OpenFlow is the first standard communications interface defined between the control and forwarding layers of an SDN architecture. It allows an external controller to interact with a switch's forwarding plane - inserting, modifying, and deleting flow rules dynamically.

Each OpenFlow-enabled switch contains:

A flow table with rules that match packets and specify actions (e.g., forward, drop, send to controller).

A secure channel to communicate with the SDN controller.

An OpenFlow protocol to manage flow table entries.

Mininet: A Lightweight SDN Testbed Mininet is a network emulator that runs a collection of virtual hosts, switches, links, and controllers on a single Linux machine. It supports:

Rapid prototyping of SDN applications

Testing controller-switch interactions

Simulating various topologies using Python or CLI

Mininet uses Open vSwitch (OVS) as the virtual switch, which is OpenFlow compatible and supports many SDN features.

How It All Works in Practice

When a host sends a packet:

The switch checks its flow table.

If no matching rule exists, it forwards the packet to the controller using OpenFlow.

The controller analyzes the packet and installs a flow rule in the switch. The switch then forwards future packets directly, reducing latency and controller load.

This behavior mimics real-world programmable networks used in data centers, enterprise networks, and cloud infrastructure.

Educational Value

Simulating OpenFlow using Mininet gives students hands-on experience with SDN principles:

Dynamic network control

Traffic engineering

Protocol interaction between switches and controllers

Real-time flow table inspection

This prepares students to work with cutting-edge network technologies like SD WAN, cloud networking, 5G, and network automation platforms.

Commands:

1. Start Mininet with 3 Hosts and 1 Switch `mn --topo single,3 --mac --switch ovsk --controller remote`

◆ 2. Start the POX Controller (in a new terminal) `cd~/pox ./pox.py forwarding.12_learning` (Alternatively, use Mininet's default controller with `--controller=default` if POX is not installed)

◆ 3. Run Basic Connectivity Tests from Mininet CLI `mininet pingall` ◆ 4. Test Bandwidth Between Two Hosts `mininet hl iperf -s & mininet h2 iperf -c hl`

◆ 5. View Flow Table Entries on the Switch `mininet sh ovs-ofctl dump-flows s1`